

- 5 (a) (i) State Kirchhoff's second law.

.....

 [1]

- (ii) State the conservation law that gives rise to Kirchhoff's second law.

..... [1]

- (b) A circuit contains a cell of internal resistance r and two resistors of resistances R_1 and R_2 , as shown in Fig. 5.1.

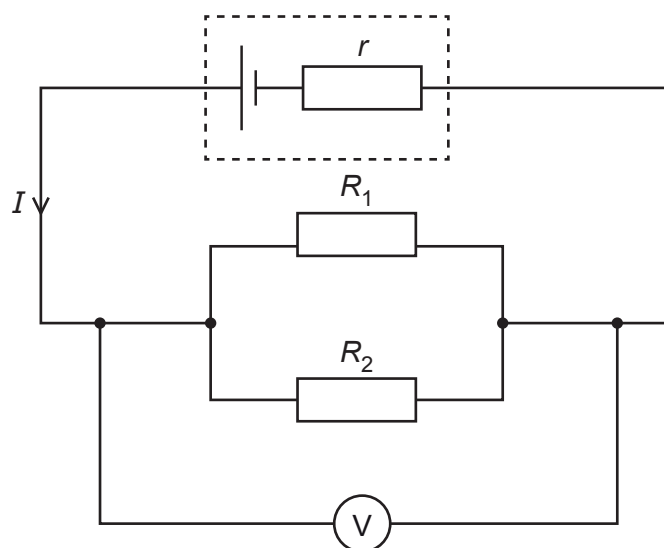


Fig. 5.1

The potential difference (p.d.) across the two resistors is V .

The current in the cell is I .

- (i) Use Kirchhoff's laws to show that the total resistance R_T of the external circuit is given by

$$\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2}.$$

[2]

- (ii) The electromotive force (e.m.f.) of the cell is 1.50 V.

When the values of R_1 and R_2 are $10\,\Omega$ and $15\,\Omega$ respectively, the p.d. measured by the voltmeter is 1.38 V.

Calculate the internal resistance r of the cell.

$r = \dots\dots\dots \Omega$ [3]

- (c) A third resistor is added in parallel with R_1 and R_2 in the circuit in Fig. 5.1.

State and explain the effect, if any, of this change on:

- (i) the current in the cell

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..... [2]

- (ii) the p.d. measured by the voltmeter.

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..... [2]

[Total: 11]